

Gavin Kerrigan

gavin.m.kerrigan@gmail.com • gavinkerrigan.github.io

Research

Postdoctoral Research Assistant in Machine Learning

Grade 8 Jan. 2026 - Present
Grade 7 Feb. 2025 - Jan. 2026
University of Oxford, Department of Statistics
RainML Lab (PI: Tom Rainforth)

Graduate Student Researcher

Sep. 2019 - Dec. 2024

University of California, Irvine, Department of Computer Science
DataLab (PI: Padhraic Smyth)

Undergraduate Research Assistant

Sep. 2018 - June 2019

The Pennsylvania State University, Department of Computer Science

Education

Ph.D., Computer Science

Sep. 2019 - Dec. 2024

University of California, Irvine GPA: 4.0/4.0
Thesis: Infinite-Dimensional Generative Models through the Transport of Measure
Advisor: Padhraic Smyth

B.Sc., Mathematics

2015 - 2019

The Pennsylvania State University, Schreyer Honors College GPA: 3.98/4.0
Minor: Physics

Teaching

Tutor

Jan. 2026 - Mar. 2026

SB2.2: Statistical Machine Learning, University of Oxford

Instructor

March 2023 - June 2023

CS 178: Machine Learning and Data Mining, UC Irvine
■ Enrollment: 162

Head Teaching Assistant

Sep. 2022 - Dec. 2022

CS 178: Machine Learning and Data Mining, UC Irvine
■ Managed a team of 5 graduate TAs in a course with an enrollment of 272
■ Developed homework assignments, weekly discussions, exams, and rubrics

Guest Lecturer

Fundamentals of AI II: Modern Statistical Concepts, University of Oxford Nov. 2025
CS 274E: Deep Generative Models, UC Irvine Nov. 2022, Nov. 2023, Nov. 2024

Private Tutor

Jul. 2019 - Dec. 2024

Machine learning, mathematics, and statistics

- Perfect (5.0/5.0) rating across more than 300 hours of tutoring and 200 reviews

Mathematics Tutor and Peer Coordinator

Jan. 2017 - June 2019

Penn State Learning

- Tutored undergraduate mathematics courses in one-on-one and small group settings
- Trained, supervised, and interviewed a team of more than 60 undergraduate tutors

Publications

Google Scholar: <https://scholar.google.com/citations?user=2F2XCy8AAAAJ>

* denotes joint authorship

Conference

1. **G. Kerrigan**, C. Naesseth, T. Rainforth, 'A Geometric Approach to Optimal Experimental Design,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, to appear, May 2026.
2. **G. Kerrigan**, K. Nelson, P. Smyth, 'EventFlow: Forecasting Continuous-Time Event Data with Flow Matching,' *Proceedings of the 29th International Conference on AI and Statistics (AISTATS 2026)*, to appear, May 2026 (**Oral, top 0.5%**).
3. A. Mammadov, J. Yao, J. Berner, **G. Kerrigan**, J. C. Ye, K. Azizzadenesheli, A. Anandkumar, 'Guided Diffusion Sampling on Function Spaces with Applications to PDEs,' *Advances in Neural Information Processing Systems 39 (NeurIPS 2025)*, December 2025.
4. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Dynamic Conditional Optimal Transport through Simulation-Free Flows,' *Advances in Neural Information Processing Systems 38 (NeurIPS 2024)*, pp.93602–93642, December 2024.
5. P. Srivastava, R. Yang, **G. Kerrigan**, G. Dresdner, J. McGibbon, C. Bretherton, S. Mandt, 'Precipitation Downscaling with Spatiotemporal Video Diffusion,' *Advances in Neural Information Processing Systems 38 (NeurIPS 2024)*, pp. 56374-56400, December 2024.
6. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *Proceedings of the 27th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 238:3934-3942, April 2024 (**Oral, Outstanding Student Paper Highlight award (7/2181)**).
7. **G. Kerrigan**, J. Ley, P. Smyth, 'Diffusion Generative Models in Infinite Dimensions,' *Proceedings of the 26th International Conference on AI and Statistics (AISTATS)*, Proceedings of Machine Learning Research, PMLR 206:9538–9563, April 2023.
8. **G. Kerrigan**, P. Smyth, and M. Steyvers, 'Combining Human Predictions with Model Probabilities via Confusion Matrices and Calibration,' *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, pp.4421–4434, December 2021.

Journal

9. C. Guilloteau, **G. Kerrigan**, K. Nelson, G. Migliorini, P. Smyth, R. Li, E. Foufoula-Georgiou, 'A Generative Diffusion Model for Probabilistic Ensembles of Precipitation Maps Conditioned

on Multisensor Satellite Observations,' *IEEE Transactions on Geoscience and Remote Sensing*, doi:10.1109/TRGS.2025.3548518, March 2025.

10. M. Steyvers, H. Tejada, **G. Kerrigan**, P. Smyth, 'Bayesian Modeling of Human-AI Complementarity,' *Proceedings of the National Academy of Sciences*, 19(11):1-7, March 2022.

Workshop

11. **G. Kerrigan**, G. Migliorini, P. Smyth, 'Functional Flow Matching,' *NeurIPS Workshop on Diffusion Models*, 2023.
12. **G. Kerrigan**^{*}, D. Slack^{*}, J. Tuyls^{*}, 'Differentially Private Language Models Benefit from Public Pre-Training,' *PrivateNLP: EMNLP Workshop on Privacy and Natural Language Processing*, pp.39-45, 2020.

Under Review

13. A. Saravanan, R. Knowles, **G. Kerrigan**, T. Rainforth, 'Scaling Bayesian Experimental Design to High-Dimensions with Information-Guided Diffusion', *under review*, October 2025.

Honors and Awards

Research

- | | |
|--|------|
| Oral Presentation (0.5% of submissions), AISTATS | 2026 |
| Outstanding Student Paper Highlight award, AISTATS | 2024 |

University of California, Irvine

- | | |
|--|-------------|
| Hasso Plattner Institute Fellowship | 2021 - 2024 |
| Graduate Dean's Recruitment Fellowship | 2019 |

The Pennsylvania State University

- | | |
|---|------------|
| Steven and Sherry McCrystal Mathematics Award | 2019 |
| Mary Lister McCammon Scholarship | 2017, 2018 |
| Phi Beta Kappa Honors Society | 2017 |
| Hay Memorial Scholarship | 2016 |
| Mahle Trustee Scholarship in Science | 2015 |

Invited Talks

Optimal Transport and Optimal Experimental Design

- | | |
|--|-----------|
| Computational Methods in Applied Mathematics (<i>Upcoming</i>) | July 2026 |
|--|-----------|

Generative Models and the Transport of Measure

- | | |
|---|--------------|
| Anandkumar Group, Caltech | August 2024 |
| University of Southern California (USC) | October 2024 |
| Yu Group, University of California San Diego (UCSD) | October 2024 |

Diffusion Generative Models: Methods and Applications

- | | |
|--------------------------------------|-----------|
| NASA Jet Propulsion Laboratory (JPL) | June 2024 |
|--------------------------------------|-----------|

Functional Flow Matching

The 27th International Conference on AI and Statistics (AISTATS)	May 2024
Deep Generative Models in Infinite-Dimensional Spaces	
Center for Machine Learning Seminar, UC Irvine	Jan. 2024
Probability and Combinatorics Seminar, UC Irvine	Nov. 2023
AMLab Seminar, University of Amsterdam	Nov. 2023
A Tutorial on Diffusion Generative Models	
Hasso Plattner Institute	Sep. 2023

Service

Departmental Committee Postdoctoral Representative

University of Oxford, Department of Statistics	Nov. 2025 - Present
--	---------------------

Student Success Mentor

The Mentor Collective, UC Irvine Next Gen Pathways	2024
--	------

Workflow Chair

AISTATS	2024
---------	------

Session Chair

AISTATS	2024
---------	------

Area Chair

ICLR, ICML	2025
------------	------

AI Faculty Search Committee, Student Member

UC Irvine	2021 - 2023
-----------	-------------

Machine Learning Hackathon Mentor

UC Irvine	2021
-----------	------

Academic Reviewing

JMLR, ICML, NeurIPS, AAAI, AABI, AISTATS, TAG-DS, The American Statistician	2025
JMLR, ICLR, CVPR, ICML, UAI, AAAI, AISTATS, NeurReps Workshop	2024
NeurIPS, AISTATS, Workshop on Deep Generative Models for Health,	2023
Workshop on Topology, Algebra, and Geometry in Data Science,	
Workshop on Diffusion Models	
ICML, Workshop on Geometrical and Topological Representation Learning,	2022
Workshop on Human-Machine Collaboration and Teaming	
Workshop on Uncertainty in Deep Learning	2020